

The Causal-Mimicry Score — Explained

How we turn a model's elicited causal-direction answers into one number per relation: how well it mimics causal reasoning, measured against a random guesser. This is the contribution — no off-the-shelf metric scores this.

1 · What we measure & why it's ours

The model is **always parroting** (obstruction theorem). So we don't ask "is it reasoning" — we ask **how well does its parroting reproduce correct causal structure**, and we score that against chance. There is no published "causal-mimicry score" to import — defining it is the novel piece. The score is computed **only from elicited behaviour vs. ground truth**, never from the internal representation metric (variable ②) — otherwise the mediation would be circular.

2 · The elicitation it scores (three-way)

Each pair is queried three-way (Long et al. 2024; Susanti & Färber 2025), context reset, sampled k times:

- (A) X causes Y — direction 1
- (B) Y causes X — direction 2
- (C) no causal relationship — correlation/none (**no direction**)

Correlation having no direction is exactly why the third option exists — and it's what lets us score whether the model distinguishes correlation from causation. On an open model we read the A/B/C token probabilities directly.

3 · Two sub-scores (kept separate)

① Direction

On genuinely causal edges: does the model orient cause → effect correctly?
Observed = mean probability on the *correct* direction token.
chance = 1/3 (three options)

② Discrimination

Across all edges: does it say "causal" when causal and "neither" when merely correlated? Observed = accuracy of the causal-vs-C call.
chance = 1/2 (causal vs not)

They measure *different* competencies — a model can orient well yet fail to tell correlation from causation, or vice versa. Averaging them into one number hides that, so we report both.

4 · Normalising against chance (the key step)

Raw accuracy misleads because **chance isn't zero**: 0.5 on a three-way task is barely above the 0.33 floor. We rescale so guessing = 0 and perfect = 1:

$$\text{normalized} = (\text{observed} - \text{chance}) / (1 - \text{chance})$$

TERM	MEANING
observed – chance	How far above guessing the model got — the improvement it actually earned.
1 – chance	How far above guessing it was <i>possible</i> to get — the room for improvement (floor to ceiling).
the ratio	What fraction of the achievable improvement the model captured. 0 = no better than guessing; 1 = perfect; <0 = worse than chance.

Why it's essential here: the two sub-scores have *different* floors (0.33 vs 0.5), so raw accuracies aren't comparable. Normalising puts both on the same "fraction of skill captured" scale, so they can share an axis in the mediation and the figure.

Worked example — direction sub-score

- Model assigns mean P(correct direction) = **0.60** across the causal edges.
- chance = 0.33, so room-for-improvement = $1 - 0.33 = 0.67$.
- $\text{normalized} = (0.60 - 0.33) / 0.67 = 0.27 / 0.67 \approx 0.40$.
- Reading: the model closed **40%** of the gap between guessing and perfect.

5 · Combining across a graph & size-invariance

For a multi-edge graph, combine per-edge scores with the **mean** (or mean-log of P-correct) — **never the raw product**. The product ($0.8 \times 0.4 \times 0.9 \dots$) shrinks toward zero with every edge, so it would conflate *graph size* with *mimicry quality*: a model equally good per-edge would look far worse on a bigger graph purely from multiplying more numbers. The mean is size-invariant — equal per-edge quality scores the same at 3 edges or 20.

The non-negotiable rule: every quantity here comes from *elicited behaviour* vs. *ground truth only* — never from variable ② (LRE representation quality). If the mimicry score depended on rep-quality, the mediation chain frequency $\rightarrow$ rep-quality $\rightarrow$ mimicry would be circular and the result meaningless. Behaviour and representation are measured from separate sources, always.

Variable ③ (γ) for the mediation pilot. Format & baselines grounded in Long et al. 2024 / Susanti & Färber 2025 (three-way pairwise). Normalisation is the standard chance-corrected rescaling (kappa-style). Implement in [mimicry/score_mimicry.py](#) against this spec.